

Telerehabilitation in patients with respiratory tract diseases: protocol for a scoping review

Shunsuke Taito^{1,2}, Kota Yamauchi³, Yuki Kataoka^{2,4,5,6}

1. Division of Rehabilitation, Department of Clinical Practice and Support, Hiroshima University Hospital, Kasumi 1-2-3, Minami-ku, Hiroshima, 734-8551 Japan
2. Systematic Review Workshop Peer Support Group (SRWS-PSG), Japan
3. Department of Rehabilitation, Steel Memorial Yawata Hospital, Fukuoka, Japan
4. Hospital Care Research Unit, Hyogo Prefectural Amagasaki General Medical Center, Higashinaniwa-cho 2-17-77, Amagasaki 660-8550 JAPAN.
5. Department of Healthcare Epidemiology, Graduate School of Medicine and Public Health, Kyoto University, Yoshida Konoe-cho, Sakyo-ku, Kyoto 606-8501 JAPAN
6. Department of Respiratory Medicine, Hyogo Prefectural Amagasaki General Medical Center, Higashinaniwa-cho 2-17-77, Amagasaki 660-8550 JAPAN

Background

Telerehabilitation is delivery of therapeutic rehabilitation at a distance or offsite using telecommunication technologies. Telerehabilitation has been proposed as a health service option to enable more patients to receive rehabilitation regardless of where they live (1). Although there was some systematic review of telerehabilitation for patients with stroke (2), multiple sclerosis (3), and lower-limb joint replacement (4), telerehabilitation for respiratory diseases and respiratory infections is not known. During the COVID-19 pandemic, telerehabilitation may represent the first option for people at home (5). The COVID-19 is one of respiratory tract disease, and rapid and widespread transmission was demonstrated (6). Therefore, it is necessary consider telerehabilitation in patients with respiratory tract disease in the acute phase in hospital as well as chronic phase from the perspective of infection prevention. The purpose of this scoping review is to assess the intervention program, safety and feasibility of telerehabilitation for patients with respiratory tract diseases in the COVID-19 era.

Methods

We describe this protocol that follows the PRISMA extension for scoping reviews (PRISMA-ScR) statement(7). We also use the scoping review framework by The Joanna Briggs Institute

(JBI) (8) following five stage approach as below.

Stage 1: Identifying the research question

We are interested in telerehabilitation for patients with respiratory tract disease, and the research questions are as followed:

1. “What is telerehabilitation for patients with respiratory tract disease?”
2. “How safe is telerehabilitation for patients with respiratory tract disease?”
3. “How feasible is telerehabilitation for patients with COVID-19?”

Stage 2: Identifying relevant studies

We used the Population, Concept, and Context (PCC) framework (8) to define the inclusion criteria as follows:

Participants

All published studies that include patients with respiratory tract disease including respiratory infection such as severe acute respiratory syndrome (SARS) coronavirus (CoV) infections, Middle East respiratory syndrome (MERS) CoV and COVID-19. In this scoping review, ICD-10 codes of the participants include J00-J99 Diseases of the respiratory system, A15-16 Respiratory tuberculosis, U04 Severe acute respiratory syndrome, U07.1 COVID-19, virus identified, U07.2 COVID-19, virus not identified.

Concept

We will review the existing literature on telerehabilitation for patients with respiratory tract disease regarding rehabilitation programs, feasibility and safety.

Context

In this review, telerehabilitation is defined as “Delivery of therapeutic rehabilitation at a distance or offsite using telecommunication technologies”. We will also include the study for telerehabilitation, remote rehabilitation and virtual rehabilitation as defined by authors. Studies in any setting (including in hospital) and phase and from any country and with any length of follow-up will be included.

The following databases will be searched: MEDLINE, Embase, the Cochrane Central Register of Controlled Trials (CENTRAL), CINAHL and PEDro. In addition, we will search for ongoing trials in the following trials register: the World Health Organization International Clinical Trials Registry Platform and ClinicalTrials.gov. We will also search for references in the articles identified for inclusion.

All published randomized controlled trials, crossover trials, cluster-randomized trials, quasi-randomized trials and non-randomized trials. observational studies with control, case reports and case series will be included. Studies in any languages and from any country will be accepted, and we will include study with any length of follow-up. We will exclude conference abstracts and review articles

Search strategy

MEDLINE search strategy

- #1 Respiratory Tract Diseases[mh]
- #2 coronavirus infections[mh]
- #3 severe acute respiratory syndrome coronavirus 2[Supplementary Concept]
- #4 bronch*[tiab]
- #5 respiratory[tiab]
- #6 tuberculosis[tiab]
- #7 "cystic fibrosis"[tiab]
- #8 pulmonary[tiab]
- #9 lung[tiab]
- #10 pneumonia[tiab]
- #11 respiration[tiab]
- #12 cough[tiab]
- #13 dyspnea[tiab]
- #14 aspiration[tiab]
- #15 tracheal[tiab]
- #16 COVID[tiab]
- #17 SARS[tiab]
- #18 MERS[tiab]
- #19 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12
OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #18
- #20 Telerehabilitation [mh]
- #21 "tele rehabilitation*" [tiab]
- #22 "remote rehabilitation*" [tiab]
- #23 "virtual rehabilitation" [tiab]
- #24 #20 OR #21 OR #22 OR #23
- #25 #19 AND #24

Embase

- S1 EMB.EXACT.EXPLODE(Respiratory Tract Diseases)

S2 EMB.EXACT.EXPLODE(coronavirus infections)
 S3 ab(severe acute respiratory syndrome coronavirus 2) OR ti(severe acute respiratory syndrome coronavirus 2)
 S4 ab(bronch*) OR ti(bronch*)
 S5 ab(respiratory) OR ti(respiratory)
 S6 ab(tuberculosis) OR ti(tuberculosis)
 S7 ab("cystic fibrosis") OR ti("cystic fibrosis")
 S8 ab(pulmonary) OR ti(pulmonary)
 S9 ab(lung) OR ti(lung)
 S10 ab(pneumonia) OR ti(pneumonia)
 S11 ab(respiration) OR ti(respiration)
 S12 ab(cough) OR ti(cough)
 S13 ab(dyspnea) OR ti(dyspnea)
 S14 ab(aspiration) OR ti(aspiration)
 S15 ab(tracheal) OR ti(tracheal)
 S16 ab(COVID) OR ti(COVID)
 S17 ab(SARS) OR ti(SARS)
 S18 ab(MERS) OR ti(MERS)
 S19 S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18
 S20 EMB.EXACT.EXPLODE(Telerehabilitation)
 S21 ab("tele rehabilitation*") OR ti("tele rehabilitation*")
 S22 ab("remote rehabilitation*") OR ti("remote rehabilitation*")
 S23 ab("virtual rehabilitation") OR ti("virtual rehabilitation")
 S24 S20 OR S21 OR S22 OR S23
 S25 S19 AND S24
 S26 (S19 AND S24) not (rtype.exact("Conference Abstract" OR "Conference Review" OR "Editorial"))

Cochrane Central Register of Controlled Trials (CENTRAL)

([mh "Respiratory Tract Diseases"] OR [mh "coronavirus infections"] OR bronch*:ti,ab OR respiratory:ti,ab OR tuberculosis:ti,ab OR "cystic fibrosis":ti,ab OR pulmonary:ti,ab OR lung:ti,ab OR pneumonia:ti,ab OR respiration:ti,ab OR cough:ti,ab OR dyspnea:ti,ab OR aspiration:ti,ab OR tracheal:ti,ab OR COVID:ti,ab OR SARS:ti,ab OR MERS:ti,ab) AND
 ([mh Telerehabilitation] OR "tele rehabilitation*":ti,ab OR "remote rehabilitation*":ti,ab OR "virtual rehabilitation":ti,ab)

CINAHL

((MH "Respiratory Tract Diseases+") OR (MH "coronavirus infections+") OR TI bronch* OR AB bronch* OR TI respiratory OR AB respiratory OR TI tuberculosis OR AB tuberculosis OR TI "cystic fibrosis" OR AB "cystic fibrosis" OR TI pulmonary OR AB pulmonary OR TI lung OR AB lung OR TI pneumonia OR AB pneumonia OR TI respiration OR AB respiration OR TI cough OR AB cough OR TI dyspnea OR AB dyspnea OR TI aspiration OR AB aspiration OR TI tracheal OR AB tracheal OR TI COVID OR AB COVID OR TI SARS OR AB SARS OR TI MERS OR AB MERS) AND ((MH "Telerehabilitation +") OR TI "tele rehabilitation*" OR AB "tele rehabilitation*" OR TI "remote rehabilitation*" OR AB "remote rehabilitation*" OR TI "virtual rehabilitation" OR AB "virtual rehabilitation") 8 件

PEDro.

Abstract & Title: telerehabilitation OR “remote rehabilitation” OR “virtual rehabilitation”

ClinicalTrials.gov

Condition or disease: Respiratory OR COVID OR MERS OR SERS OR tuberculosis
OR pulmonary

Intervention: Telerehabilitation OR tele rehabilitation OR remote rehabilitation OR
virtual rehabilitation

Stage 3: Study selection

We will select studies following the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) flow diagram (9). We will use the PCC framework and the process will be carried out independently by two researchers (ST and KY). The two authors will compare their lists and any differences in opinion will be resolved by discussion and, where this fails, through arbitration by a third researcher (YK).

Stage 4: Charting the data

Data extraction will be carried out by one researcher (ST) using standard data extraction forms including diagnosis, setting, study type, number of participants, telerehabilitation program, control, outcomes, adverse events, and feasibility. Another one researcher (KY) confirm the data extraction. If necessary, we will contact the authors of these studies.

Stage 5: Collating, summarizing, and reporting the results

We will present the results of the search in the PRISMA flow diagram. We will organize the extracted data described above as qualitative synthesis (Table 1, Table 2).

Table 1. Telerehabilitation for patients with respiratory tract disease

Author, year, country	Diagnosis	Setting	Study type	Number of participant	Telerehabilitation program (frequency, intensity, time, type)	Control	Outcomes

Table 2. Safety and feasibility of telerehabilitation for patients with respiratory tract disease

Author, year, country	Diagnosis	Adverse event	Withdrawals	Adherence	Face-to-face rehabilitation evaluation	Face-to-face rehabilitation session	Ratio of telerehabilitation to all rehabilitation session

Differences between the protocol and the overview

Changes in the methodology that need to be carried out throughout the study will be described in the results section.

Ethics and dissemination

In this scoping review, only data from previously conducted studies will be analyzed. We will publish this scoping review in peer-reviewed journals.

Patients and or public involvement

Patients and or public were not involved in the development of this protocol. The study group developed this study protocol without patient involvement.

Contribution

ST conceived the study. ST, KY and YK contributed to the protocol design and plan. ST and KY will develop the literature search strategy and charting data. ST, KY and YK will draft and revise manuscript.

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